

HW 1

Problem 1

Let G be a group and $N \trianglelefteq G$

Let ψ be an homomorphism from G to G_1 s.t. $N \subseteq \ker(\psi)$

Define: $\bar{\psi} : G/N \rightarrow G_1$ s.t. $\bar{\psi}(gN) = \psi(g)$

Well Defined: Suppose $gN = hN \implies gh^{-1} \in N \implies gh^{-1} \in \ker(\psi)$

$\psi(gh^{-1}) = e \implies \psi(g) = \psi(h)$ since ψ is a homomorphism this implies $\bar{\psi}(gN) = \bar{\psi}(hN)$

Homomorphism: Let $gN, hN \in G/N$

$\bar{\psi}(gNhN) = \bar{\psi}(ghN) = \psi(gh) = \psi(g)\psi(h) = \bar{\psi}(gN)\bar{\psi}(hN)$

Uniqueness: Suppose there exists another homomorphism $\phi : G/N \rightarrow G_1$ such that $\phi = \psi \circ \varphi$

$$\phi(gN) = \psi(g) \implies \phi = \bar{\psi}$$

Problem 2

Let H be a subgroup of G .

- (Inverses) Let $a \in N_G(H)$
Consider: $a^{-1}Ha = a^{-1}aH = H$ thus $a^{-1} \in N_G(H)$
- (Closure) Let $a, b \in N_G(H)$
Consider: $abHb^{-1}a^{-1} = abb^{-1}Ha^{-1} = aHa^{-1} = H$ thus $ab \in N_G(H)$
- (Identity) Consider $eHe = H$ thus $e \in N_G(H)$

Let $h \in H$

Consider : hHh^{-1} since $h, h^{-1} \in H \implies hHh^{-1} = H$ thus $h \in N_G(H)$ thus $H \subseteq N_G(H)$

By definition since $\forall g \in N_G(H) gHg^{-1} = H$ thus $H \trianglelefteq N_G(H)$

Largest: Let K be a subgroup of G s.t. $H \trianglelefteq K$ thus $\forall k \in K$

$$kHk^{-1} = H$$

thus $k \in N_G(H)$ thus $K \subseteq N_G(H)$

Problem 3

- (i)

Proof.

- (Closure) Let $hk, jl \in HK$ s.t. $h, j \in H$ and $k, l \in K$
 $hkjl = h(kj)l$ since $H \subseteq N_G(K) \implies kj = jk'$ for $k' \in K$
 $h(jk')l = hjk'l \in HK$
- (Identity) Since H, K are subgroups of G , $e \in H \cap K \implies ee = e \in HK$
- (Inverses) Let $hk \in HK$ s.t. $h \in H$ and $k \in K$
 Consider $k^{-1}h^{-1}$ since $H \subseteq N_G(K) \implies k^{-1}h^{-1} = h^{-1}k'$ for $k' \in K$
 thus $k^{-1}h^{-1} = h^{-1}k' \in HK$
- Let $hk \in HK$ s.t. $h \in H$ and $k, k' \in K$
 $(hk)k'(hk)^{-1} = h(kk'k^{-1})h^{-1}$ thus $kk'k^{-1} = l \in K$ and since $h \in H \subseteq N_G(K)$
 $\implies hllh^{-1} \in K$ thus $(hk)k'(hk)^{-1} \in K$ thus $K \trianglelefteq HK$

□

- (ii)

Proof. Let $l \in H \cap K \implies l \in H$ and $l \in K$ and $h \in H$
 hlh^{-1} since $l \in K$ and $H \subset N_G(K) \implies hllh^{-1} \in K$
 since $h \in H$ and $l \in H \implies hllh^{-1} \in H$ thus $hllh^{-1} \in H \cap K$ thus $H \cap K \trianglelefteq H$

□

- (iii)

Proof. Let $\varphi : H \rightarrow HK/K$ s.t. $\varphi(h) = hK$
 Well-defined: Suppose $h_1 = h_2 \in H \implies \varphi(h_1) = h_1K = h_2K = \varphi(h_2)$
 Homomorphism: Let $h, j \in H$
 $\varphi(hj) = hjK = hKjK = \varphi(h)\varphi(j)$
 Kernel: Let $h \in \ker(\varphi) \trianglelefteq H \implies h \in H$
 $\varphi(h) = hK = K \implies h \in K \implies h \in H \cap K \implies \ker(\varphi) = H \cap K$
 Surjective: Let $hkK \in HK/K$, for $h \in H, k \in K$
 $\varphi(h) = hK = hkK$ thus $\text{im}(\varphi) = HK/K$
 Thus by the first isomorphism theorem $H/H \cap K \cong HK/K$

□

Problem 4

- (a) • (i. $\implies$ ii.)

Assume i.

Let $g \in G$ by i. we know $g = hk$ for $h \in H$ and $k \in K$ thus $g = hk \in HK$

Let $hk \in HK$ s.t. $h \in H$ and $k \in K$ since $H, K \subseteq G \implies hk \in G$ thus $G = HK$

Since $e \in H, K \implies e \in H \cap K \implies \{e\} \subseteq H \cap K$

Let $l \in H \cap K \implies l \in H$ and $l \in K$ and $g \in G$ by i. $g = hk$ for $h \in H$ and $k \in K$

$g = hxx^{-1}k$ thus $hx \in H$ and $x^{-1}k \in K$ thus by i. since we assume uniqueness for each $g \in G$ this implies $x = e$ thus $H \cap K = \{e\}$

- (ii. $\implies$ iii.)

Assume ii.

Define: $\varphi : H \times K \rightarrow G$ s.t. $\varphi(h, k) = hk$

Homomorphism: Let $(h, k), (j, l) \in H \times K$

$\varphi((h, k)(j, l)) = \varphi(hj, kl) = hjkl = \varphi(hj)\varphi(kl)$

Injective: Let $(h, k) \in \ker(\varphi)$

$\varphi(h, k) = hk = e \implies h = k^{-1} \implies h, k \in H \cap K$ by ii. $\implies h = k = e$

Thus $(h, k) = (e, e) \implies \ker(\varphi) = \{(e, e)\} \implies \varphi$ is injective.

Surjective: Let $g \in G$ by ii. $g = hk$ for $h \in H$ and $k \in K$ thus $\varphi(h, k) = hk = g$ thus surjective

Thus φ is an isomorphism

- (iii. $\implies$ i.)

Assume iii.

Let $g \in G$ and $k \in K$

By iii. since $H \times K$ is isomorphic to G this implies for all $g \in G$ can be written uniquely as a product hk , with $h \in H$ and $k \in K$

Let $g \in G$ and $h_1 \in H$

gh_1g^{-1} by iii. $g = hk$ for $h \in H$ and $k \in K$

$hkh_1k^{-1}h^{-1}$

Consider $kh_1k^{-1} \in G \implies kh_1k^{-1} = k\varphi(h_1, k^{-1}) = k\varphi((e, k^{-1})(h_1, e)) = k\varphi(e, k^{-1})\varphi(h_1, e) = kk^{-1}h_1 = h_1 \in H$

Thus $kh_1k^{-1} = h_1 \in H \implies gh_1g^{-1} \in H$ thus H is normal in G

Let $g \in G$ and $k_1 \in K$

gk_1g^{-1} by iii. $g = hk$ for $h \in H$ and $k \in K$

$hkk_1k^{-1}h^{-1}$. Let $l = kk_1k^{-1}$

$hll^{-1} = \varphi(h, l)h^{-1} = \varphi(e, l)\varphi(h, e)h^{-1} = lhh^{-1} = l \in K \implies K \trianglelefteq G$

- (b) If G is an internal direct product of H and K then $G \cong H \times K$

Problem 5

- (a) $\varphi : \{\text{subgroups of } G \text{ containing } N\} \rightarrow \{\text{subgroups of } G/N\}$ s.t. $\varphi(H) = \phi(H) = H/N$
 $\varphi^{-1} : \{\text{subgroups of } G/N\} \rightarrow \{\text{subgroups of } G \text{ containing } N\}$ s.t. $\varphi^{-1}(\bar{H}) = \phi^{-1}(\bar{H})$

Let $H \leq G$

$$\varphi^{-1}(\varphi(H)) = \varphi^{-1}(H/N) = H$$

Let $\bar{H} \leq G/N$

$$\varphi(\varphi^{-1}(\bar{H})) = \varphi(\phi(\bar{H})) = \bar{H}$$

Thus φ is bijective since it has an inverse

Let $H \trianglelefteq G$ and $H \supseteq N$ consider $\varphi(H) = H/N$

Let $gN \in G/N$ and $hN \in H/N$

$$gN h N g^{-1} N = g h g^{-1} N \text{ since } H \trianglelefteq G \implies g h g^{-1} \in H \implies g N h N g^{-1} N = g h g^{-1} N \in H/N$$

Let $\bar{H} \trianglelefteq G/N$ consider $\varphi^{-1}(\bar{H})$

Let $g \in G$ and $h \in \varphi^{-1}(\bar{H})$

$$\phi(g h g^{-1}) = g h g^{-1} N \in \bar{H} \text{ since } \bar{H} \text{ is normal} \implies g h g^{-1} \in \varphi^{-1}(\bar{H}) \implies \varphi^{-1}(\bar{H}) \trianglelefteq G$$

- (b) Let $h \in H$ and $n \in N$

$$h n h^{-1} \text{ since } h \in H \subseteq G \text{ and } N \trianglelefteq G \implies h n h^{-1} \in N \implies N \trianglelefteq H$$

By (a) since $H \trianglelefteq G \implies H/N \trianglelefteq G/N$

Let $\varphi : G/N \rightarrow G/H$ s.t. $\varphi(gN) = gH$

Well Defined: Let $xN = yN \implies xy^{-1} \in N \subseteq H$

$$\implies xy^{-1}H = H \implies xH = yH \implies \varphi(xN) = \varphi(yN)$$

Kernel: Let $gN \in \ker(\varphi)$ s.t. $g \in G$

$$\varphi(gN) = gH = H \implies g \in H \implies gN \in H/N$$

Let $hN \in H/N$ s.t. $h \in H$

$$\varphi(hN) = hH = H \implies hN \in \ker(\varphi)$$

Thus $\ker(\varphi) = H/N$

Surjective: Let $gH \in G/H$ s.t. $g \in G$

Consider: $\varphi(gN) = gH$ thus surjective

By the first isomorphism theorem $\frac{G/N}{H/N} \cong G/H$

Problem 6

Proof. If $X, Y \trianglelefteq G$ then $XY \leq G$

- (Closure) Let $a, b \in XY$ thus $a = xy$ and $b = ij$ for $x, i \in X$ and $yj \in Y$
 $ab = xyij = x(yi)j$ since X is normal in G we know $yiy^{-1} = x' \in X \implies yi = x'y$
 $= xx'yj \in XY$
- (Inverses) Let $a \in XY$ thus $a = xy$ for $x \in X$ and $y \in Y$
 $a = xy \implies a^{-1} = y^{-1}x^{-1}$ since X is normal in G we know $y^{-1}x^{-1}y = x' \in X \implies y^{-1}x^{-1} = x'y^{-1}$
 $a^{-1} = x'y^{-1} \in XY$
- (Identity) $e \in X \cap Y$ thus $ee = e \in XY$

Thus $XY \leq G$ □

Let G be order 8.

By lagrange's theorem G can only have terms of order 1,2,4,8

1. If G has an element $g \in G$ s.t. order of g is 8.
 $\langle g \rangle \leq G$ but since $|\langle g \rangle| = |G| \implies \langle g \rangle = G \implies G$ is cyclic thus $G \cong C_8$
2. If G does not have an element of order 8 or 4 thus all nonidentity elements have order 2.
 Let $a \neq b \neq e$ s.t. $a, b \in G$ thus let $X = \{e, a\}$ and $Y = \{e, b\}$
 $XY = \{e, a, b, ab\}$ since XY has 4 elements and is obviously not cyclic $XY \cong C_2 \times C_2$ and $XY \leq G$
 Let $c \notin XY$ this is possible since there are 4 elements in XY , so there are 4 remaining elements in G thus $Z = \{e, c\}$
 Consider: XYZ since G is abelian all subgroups are normal so XY and Z are normal in G and $XY \cap Z = \{e\}$
 XYZ is a subgroup of G since XY and Z are normal, so its order must divide 8 by lagrange's theorem.
 XYZ must have $\{e, a, b, ab, c\}$ so it must have at least 5 elements, since its order must divide 8 this implies its order is 8 thus $XYZ = G$ and since Z is order 2, $Z \cong C_2$
 thus by problem 4 $G = XYZ \cong XY \times Z \cong C_2 \times C_2 \times C_2$
3. If G does not have an element of order 8 thus G only has elements of order 1,2,4
 Let $a \in G$ s.t. $\text{ord}(a) = 4$ and $X = \{e, a, a^2, a^3\} = \langle a \rangle$ and let $b \notin X$
 (a) If $\text{ord}(b) = 2$ then let $Y = \{e, b\} = \langle b \rangle$
 Then X, Y are both normal since they are subgroups of an abelian group and $X \cap Y = \{e\}$

Also consider XY , XY is a subgroup of G since X and Y are normal. XY must also contain $\{e, a, a^2, a^3, b\}$ so a total of 5 elements however the order must divide the order of G so this implies $|XY| = 8 \implies XY = G$ thus by problem 4, $G \cong X \times Y$ and since X is cyclic $X \cong C_4$ and since Y is cyclic $Y \cong C_2$ thus $G \cong X \times Y \cong C_4 \times C_2$

(b) If $\text{ord}(b) = 4 \implies \text{ord}(b^2) = 2$

i. If $b^2 \notin X$ then done by proof above in case (a)

ii. If $b^2 \in H \implies b^2 = a^2$ since these have the same order.

Thus consider ab , ab is not in X since $a \in X$ and $b \notin X$

$abab = a^2b^2 = a^2a^2 = a^4 = e$ thus ab is order 2 and not in X thus done by proof above in case (a)

Problem 7

Proof. Let G be a group and $H \leq G$ and $[G : H] = 2$ then $H \trianglelefteq G$

Since $[G : H] = 2$ this implies $G = gH \sqcup H$ for $g \notin H$

Consider Hg since $g \notin H \implies Hg \neq H$

If $x \in H \implies xHx^{-1} = Hx^{-1} = H$

Thus $H \trianglelefteq G$ □

(a) If G had an element of order 8 then it would be cyclic and thus abelian

If G did not have an element of order 8 or 4 then it would only have elements of order 1,2 thus proven in class since all nonidentity elements have order 2 this makes G abelian

So G must have an element x of order 4

By lagrange's theorem, $[G : \langle x \rangle] = 2$ thus by proof above $\langle x \rangle \trianglelefteq G$ thus $xyx^{-1} \in \langle x \rangle$

Since x is order 4 this means xyx^{-1} is order 4 this $xyx^{-1} = x$ or x^3 if it is x then this implies $yx = xy$ which implies G is abelian so this is not possible thus $xyx^{-1} = x^3$

Same for $y^{-1}xy$, so it also equals x^3

(b) 1. If $\text{ord}(y) = 2$ then $G = \langle x, y \mid x^4 = y^2 = e, yx = x^3y \rangle$

Let $\varphi : G \rightarrow D_4$ s.t. $\varphi(x) = a$ where a is the rotation and $\varphi(y) = b$ where b is the reflection.

Homomorphism: Since x, y are generators only have to check $\varphi(y)\varphi(x) = \varphi(x)^3\varphi(y)$ holds and that we send same order to same order but by definition of φ we do send same order to same order

$$\varphi(y)\varphi(x) = ba = a^3b = \varphi(x)^3\varphi(y)$$

Injective:

$$\varphi(x^2) = a^2$$

$$\varphi(y^2) = b^2$$

$$\varphi(yx) = \varphi(y)\varphi(x) = ba$$

$$\varphi(yx^2) = ba^2$$

$$\varphi(yx^3) = ba^3$$

Thus injective

Surjective: Let $g \in D_4$

Thus $g = b^j a^i$ for $i \in \{0, 1, 2, 3\}$ and $j \in \{0, 1, 2, 3\}$

Consider $y^j x^i$

$$\varphi(y^j x^i) = \varphi(y^j)\varphi(x^i) = \varphi(y)^j \varphi(x)^i = b^j a^i \text{ thus surjective}$$

Thus φ is an isomorphism thus $G \cong D_4$

2. If $\text{ord}(y) = 4$ then $\text{ord}(y^2) = 2$

$$yxyx = yx^3x = y^2, \text{ so order is more than 2}$$

$$yx^2yx^2 = yx^4 = y^2, \text{ so order is more than 2}$$

$$yx^3yx^3 = yx^4 = y^2, \text{ so order is more than 2}$$

thus the only element with order 2 is x^2 thus $x^2 = y^2$

Let $\varphi : G \rightarrow Q$ s.t. $\varphi(x) = i$ and $\varphi(y) = j$

Homomorphism: Since x, y are generators only have to check order, which is the same, and

that $\varphi(y)\varphi(x)\varphi(y)^{-1} = \varphi(x)^3$ and $\varphi(y)^2 = \varphi(x)^2$

$$\varphi(y)\varphi(x)\varphi(y)^{-1} = j i j^{-1} = j i j^3 = j k j^2 = i i^2 = i^3 = \varphi(x)^3$$

$$\varphi(y)^2 = j^2 = i^2 = \varphi(x)^2$$

Injective:

$$\varphi(x^2) = \varphi(x)^2 = i^2 = -1$$

$$\varphi(y^2) = \varphi(y)^2 = \varphi(x)^2 = -1$$

$$\varphi(yx) = \varphi(y)\varphi(x) = j i = -k$$

$$\varphi(yx^2) = j - 1 = -j$$

$$\varphi(yx^3) = j(-i) = k$$

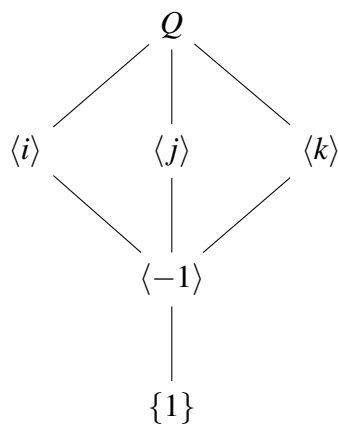
Thus injective

Surjective: Let $g \in Q$ thus $g = j^a i^b$ for $a \in \{0, 1, 2, 3\}$ and $b \in \{0, 1\}$

Consider $y^a x^b$

$$\varphi(y^a x^b) = \varphi(y^a)\varphi(x^b) = \varphi(y)^a \varphi(x)^b = j^a i^b$$

Thus φ is an isomorphism thus $G \cong Q$



(c)

Let H be a subgroup of Q by Lagrange's Theorem the order of H can only be 1, 2, 4, 8

Case 1: H is order 1

- The only subgroup of order 1 is the trivial subgroup $\{e\}$.
- Since $\{e\}$ is already on our list, H cannot be of order 1 unless it is $\{e\}$, which is already accounted for.

Case 2: H is order 2

- If H is order 2, it must consist of e and one other element g , where $g^2 = e$.

- In Q the elements with order 2 are only -1
Therefore, any subgroup H of order 2 must be $\{1, -1\}$.

Case 3: H is order 4

- If $H \cong C_4 \implies H$ has an element of order 4.
Thus, elements of order 4 are $i, -i, j, -j, k, -k$ and since
- $\langle i \rangle = \langle -i \rangle$
- $\langle j \rangle = \langle -j \rangle$
- $\langle k \rangle = \langle -k \rangle$
Thus H is one of these
- If $H \cong C_2 \times C_2 \implies H$ has only elements of order 2.
However, there are not enough elements of order 2 for this to happen.

Case 4: H is order 8

- If H is order 8, it must be the whole group Q .
- Therefore, H would be Q , which is already on our list.

In each case, we've shown that any possible subgroup H of Q must be one of the subgroups already listed. Therefore, no additional subgroups can exist, and our list of subgroups is complete.

By the proof below since $Z(Q) = \langle -1 \rangle$ this means it is normal and $\{e\}$ and the full group are always normal.

For $\langle i \rangle$ we only have to check for conjugacy with the generators

- For i we know that conjugates with all elements in $\langle i \rangle$ since subgroup closure.
- For j , $jij^{-1} = ji(-j) = (-j)ij = j^{-1}ij = kj = -i \in \langle i \rangle$
- For k , $kik^{-1} = ki(-k) = (-k)ik = k^{-1}ik = -ijij = -ij(-1)j = -(-i)j^2 = -i \in \langle i \rangle$

Thus $\langle i \rangle$ is normal and similar proof for $\langle j \rangle$ and $\langle k \rangle$ thus all subgroups of Q are normal.

- $1 \in Z(Q)$
- -1 , let $x \in Q$ $(-1)x = x(-1) = -x$ thus $-1 \in Z(Q)$
- $i, ij = k$ and $ji = -k$
- $-i, (-i)j = -k$ and $j(-i) = k$

- $j, ij = k$ and $ji = -k$
- $-j, i(-j) = -k$ and $(-j)i = k$
- $k, ki = j$ and $ik = -j$
- $-k, -ki = -j$ and $i(-k) = j$
- Thus $Z(Q) = \{1, -1\}$

Problem 8

- (Closure) Let $A, B \in H$

$$A = \begin{pmatrix} 1 & a_1 & b_1 \\ 0 & 1 & c_1 \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & a_2 & b_2 \\ 0 & 1 & c_2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 & a_1 & b_1 \\ 0 & 1 & c_1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a_2 & b_2 \\ 0 & 1 & c_2 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a_1 + a_2 & b_1 + a_1 c_2 + b_2 \\ 0 & 1 & c_1 + c_2 \\ 0 & 0 & 1 \end{pmatrix}$$

thus $AB \in H$

- (Identity) $I \in H$
- (Inverses) Let $A \in H$

$$A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} 1 & -a & ac - b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix}$$

$$AA^{-1} = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -a & ac - b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a - a & b - ac + ac - b \\ 0 & 1 & c - c \\ 0 & 0 & 1 \end{pmatrix} = I$$

$$A^{-1}A = \begin{pmatrix} 1 & -a & ac - b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a - a & b - ac + ac - b \\ 0 & 1 & c - c \\ 0 & 0 & 1 \end{pmatrix} = I$$

Thus H is a subgroup

Let $A \in H$ and $B \in K$

$$A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & g \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$ABA^{-1} = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & g \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -a & ac - b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a & b + g \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -a & ac - b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & g \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Thus $ABA^{-1} \in K$ thus $K \trianglelefteq H$

Let $h \in H$, consider hK

$$hK = \left\{ \begin{pmatrix} 1 & x & y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & g \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x & g+y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} : g \in \mathbb{R} \right\}$$

Thus $A = B$ for $A, B \in H/K$ if $a_{12} = b_{12}$ and $a_{23} = b_{23}$, a_{ij} is the (i, j) component.

Thus

$$H/K = \left\{ \left\{ \begin{pmatrix} 1 & x & g \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} : g \in \mathbb{R} \right\} : x, z \in \mathbb{R} \right\}$$

Thus let $\varphi : H/K \rightarrow \mathbb{R}^2$ s.t. for $A \in H/K$ and $x, z \in \mathbb{R}$

$$A = \left\{ \begin{pmatrix} 1 & x & g \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} : g \in \mathbb{R} \right\}$$

$$\varphi(A) = (x, z)$$

Homomorphism: Let $A, B \in H/K$ s.t.

$$A = \left\{ \begin{pmatrix} 1 & x & g \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} : g \in \mathbb{R} \right\} \quad B = \left\{ \begin{pmatrix} 1 & a & g \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : g \in \mathbb{R} \right\}$$

for $x, z, a, b \in \mathbb{R}$

$$AB = \left\{ \begin{pmatrix} 1 & x+a & g \\ 0 & 1 & z+b \\ 0 & 0 & 1 \end{pmatrix} : g \in \mathbb{R} \right\}$$

$$\varphi(AB) = (x+a, z+b) = (x, z) + (a, b) = \varphi(A)\varphi(B)$$

Injective: Let $A \in \ker(H/K)$ for $x, z \in \mathbb{R}$

$$A = \left\{ \begin{pmatrix} 1 & x & g \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} : g \in \mathbb{R} \right\}$$

$\varphi(A) = (x, y) = (0, 0) \implies x = 0$ and $y = 0$ thus

$$K = A = \left\{ \begin{pmatrix} 1 & 0 & g \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : g \in \mathbb{R} \right\}$$

thus the kernel is trivial thus φ is injective

Surjective: Let $(x, z) \in \mathbb{R}^2$ s.t. $x \in \mathbb{R}$ and $z \in \mathbb{R}$

Consider:

$$A = \left\{ \begin{pmatrix} 1 & x & g \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} : g \in \mathbb{R} \right\}$$

$\varphi(A) = (x, z)$ thus surjective

Thus $H/K \cong \mathbb{R}^2$

Let $A \in H$ and $B \in Z(H)$

$$A = \begin{pmatrix} 1 & x & y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & x' & y' \\ 0 & 1 & z' \\ 0 & 0 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 & x & y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & x' & y' \\ 0 & 1 & z' \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x+x' & y+x'z+y' \\ 0 & 1 & z+z' \\ 0 & 0 & 1 \end{pmatrix}$$

$$BA = \begin{pmatrix} 1 & x' & y' \\ 0 & 1 & z' \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & x & y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x+x' & y'+xz'+y \\ 0 & 1 & z+z' \\ 0 & 0 & 1 \end{pmatrix}$$

since $B \in Z(H) \implies AB = BA$ thus $y + x'z + y' = y' + xz' + y \implies x'z = xz' \implies x'z - xz' = 0$

1. If $x' = 0$ then $xz' = 0 \implies z' = 0$ since x could be anything
2. If $z' = 0$ then similar reasoning to above $\implies x' = 0$
3. If $x' \neq 0 \implies z' \neq 0$ thus $\frac{z'}{x'} = \frac{x}{x'}$ however since this implies that z' and x' depend on what z, x are this solution will not work.

Thus, $B \in K$

Show above for $A \in H$ and $B \in K$, $ABA^{-1} = B \implies AB = BA$ thus $B \in Z(H)$

Thus $Z(G) = K$